[536709] Stochastic Processes

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0 Introduction

This course is an introductory course on stochastic processes, and we mainly focus on intuitive and probabilistic aspects.

0.1 What is the stochastic process?

We have several ways to understand *stochastic processes*. Technically, we can describe stochastic processes from several complementary viewpoints. For convenience, we roughly classify them into three:

1. (Intuitive) **Continuum limit** of the sequence of random variables:

$$\text{Discrete process } (X_n)_{n\in\mathbb{N}} \longrightarrow \text{ Stochastic process } (X_t)_{t\geq 0}$$

2. (Structured) Infinitesimal **Generator**

$$Lf(x) = \lim_{t\downarrow 0} \frac{\mathbb{E}_x[f(X_t)] - f(x)}{t}$$

3. (Coherent) Markov **Semigroup**

$$(T_t f)(x) := \mathbb{E}_x[f(X_t)]$$

In all these viewpoints, the underlying object is the same, but the aspect we focus on differs. When we understand stochastic processes as the continuum limit of discrete processes, we mainly look at **path** of the process. If we see stochastic processes from the view of generator, we pay attention to the **operator** to functions. At the third stage, we treat stochastic processes as **time evolution**.

Continuum limit The first stage needs Probability Theory as the basic knowledge. The strong point of this direction is that we can catch the concepts on stochastic processes *intuitively*. We practically construct the process from the sequence of random variables, so it is relatively easy to understand what we are doing. From the side of application, this way of understanding is compatible with numerical simulations, such as Markov Chain Monte Carlo method. The demerit mainly comes from "limit". We first need to check the existence and uniqueness of limits. Then we have to learn tightness, Skorokhod space, embedding, or so. When we consider a continuum limit, we should care the scaling but it depends on the detail of models, which implies that techniques might be different in each model and it is not easy to construct a general theory. Even when we treat infinite dimensions or continuous time, an explosion will appear easily, which makes it difficult to analyze path regularity.

Generator The view in the second stage is more advanced. It is known that the Laplacian is deeply related to the Brownian motion. This is the most basic example but the simplest one to understand stochastic process from the view of generator. If we consider jump process or diffusion process, then we can see more general second-order differential operators or non-local operators, like fractional Laplacian. The merit of this analysis is clear connection to partial differential equations (PDE), especially elliptic/parabolic theory and analysis of Fokker–Planck-type equations. Via the generator point of view we can clearly see drift, diffusion and jump structures. Unfortunately we cannot have sample paths in the analysis and it becomes complicated to see the connection to probability theory intuitively. We need to care the domain and boundary condition. In this course, we only give a brief introduction to this viewpoint.

Semigroup In the third stage, we define an operator $(T_t f)(x) := \mathbb{E}_x[f(X_t)]$ depending on time. It is well-known that the operator is positive and has a semigroup structure. The analysis from semigroup theory is quite strong and unified, which means that discrete/continuous, jump/diffusion or finite/infinite states can be argued in the same framework via spectral theory. The theory is well-organized and we have a lot of sharp theorems, like Hille–Yoshida or Krein–Rutman. The difficulty of this analysis is that it is quite abstract and we need to learn functional analysis and operator theory. Of course it is not easy to see the connection to sample path or probabilistic intuitions. We do not develop this theory in detail, but mention it to show a unified perspective.

Other views We also have other views to stochastic processes. We can also consider stochastic processes as probability measures on path space, which can make a good connection to theory on large deviations. If you are interested in mathematical finance, you regard stochastic process as a flow of information and analyze stochastic differential equations (SDE). Stochastic control is also a hot topic.

0.2 In this course

In this introductory course on stochastic processes, we mainly adopt the first viewpoint, namely stochastic processes as continuum limits of sequences of random variables. This viewpoint is the most intuitive and suitable for a first encounter with stochastic processes.

We first review several concepts from Probability Theory and Advanced Probability Theory. Due to time constraints, we cannot take time for basic part of probability theory. So we recommend to have some references or textbooks on probability theory to check basics by yourself. If time permits, we illustrate the relationship between these viewpoints using Brownian motion, which serves as a canonical example connecting discrete limits, generators, and semigroups. We again emphasize that other viewpoints, such as generators and semigroups, will become indispensable in more advanced studies.

Contents We will separate lectures into several blocks:

I. Review of Probability Theory

II. Stochastic Processes on Discrete Settings

III. Stochastic Processes on Continuous Settings

IV. Generator and Semigroup

In the second section, we mainly consider the theory around martingales and Markov chains. In the third part, we focus on Brownian motion with martingale and Markov property. Unfortunately, lecture note is not completed, so the plan of lectures may be changed.

Evaluation In this course, we evaluate your grade by attendance, report and examination. In the lecture note, we have several exercise questions. We will treat some of them during the lectures. For examination, we will take questions we mentioned during lectures, and for report, we choose 2 or 3 questions from others.

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